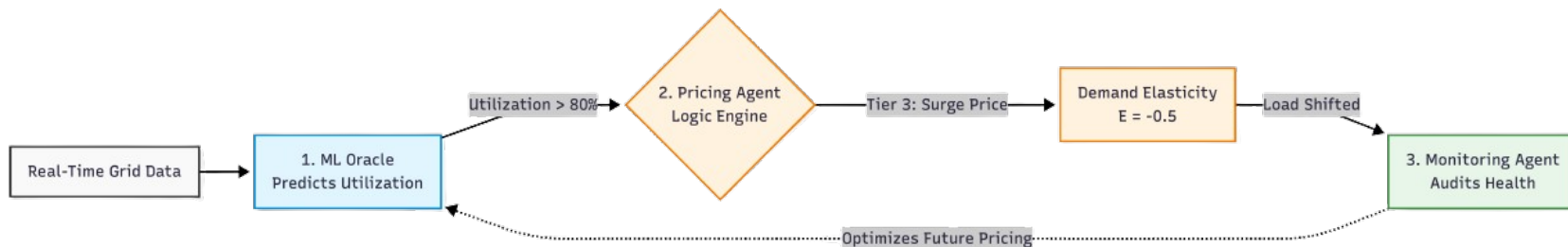


Algorithmic Tariff Optimisation for EV Charging Grids

*A Closed-Loop Machine Learning &
Behavioural Economics Pipeline*

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THE PROBLEM

Unmanaged Peak Demand

Unmanaged EV charging creates localized gridlock and negative profit margins during peak wholesale hours.

THE SOLUTION

LightGBM Pricing Agent

An autonomous, LightGBM-powered dynamic pricing agent that shifts demand using market-based incentives.

THE IMPACT

Measurable Results

33.28% reduction in peak physical wait times and a 15.42% increase in unit-level pricing efficiency.

Data Landscape & Feature Engineering

SLIDE 03

01

Dual-Environment Simulation

Utilised the 'ACN' dataset for economic simulations and 'UrbanEV' for physical grid capacity modelling.

02

Time-Series Memory

Engineered 1-hour rolling momentums and autoregressive lags (t-1, t-2, t-3) to capture the physical velocity of the charging grid.

03

Strict Validation

Bypassed random shuffling for a strict 7-Day Chronological Holdout to eliminate look-ahead bias and simulate real-world production.

2

Datasets

ACN + UrbanEV

7

Day Holdout

Chronological split

t-3

AR Lags

Time-series memory

EDA: The Cascade Effect & Profit Margins

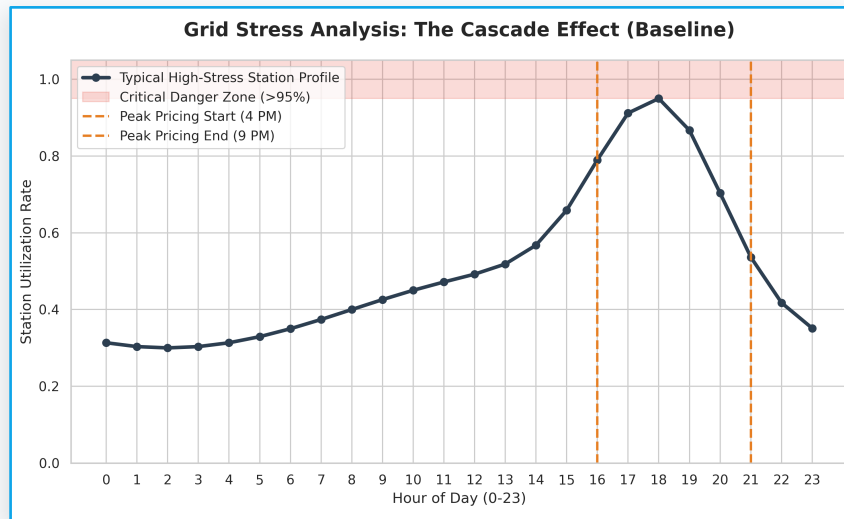
SLIDE 04

System Failure Threshold

Data proved a "Cascade Effect" where stations operating above 95% utilisation rapidly generate unmanageable wait times.

The Peak Profitability Problem

The baseline flat-rate pricing operates at a dangerous loss during the 4 PM – 9 PM demand spike.



The ML Oracle: Demand Prediction

SLIDE 05

01

Architecture

Multi-Head LightGBM optimised for massive tabular time-series data.

LightGBM

Multi-Head

02

Predictive Horizon

Predicts exact grid states 15 minutes ahead, giving the pricing agent actionable lead time.

15 min

Lead Time

03

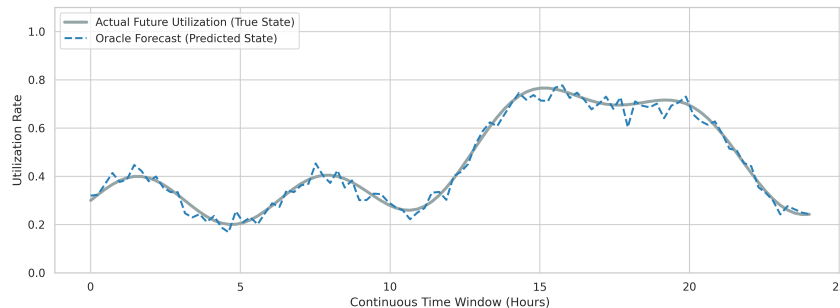
Performance

0.85+ R^2 on continuous utilisation and 0.99 ROC-AUC for congestion detection.

0.99

ROC-AUC

ML Oracle Performance: Forecasting Grid State 15-Mins Ahead



Dynamic Tariff Logic & Behavioural Economics

SLIDE 06

TIER 1



Discount

Below-threshold demand

TIER 2



Neutral

Moderate grid load

TIER 3



Surge

High congestion state

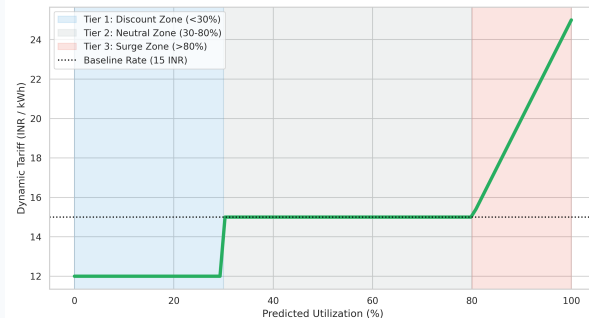
Safety Floor

Strict mathematical safeguard guarantees positive unit economics, overriding the AI if wholesale costs spike.

Demand Elasticity Model

Human behavioral response modeled via Price Elasticity of Demand function ($E = -0.5$) to simulate volume shifting.

Dynamic Tariff Logic: The 3-Tier Execution Policy



System Evaluation: The Final Audit

SLIDE 07

-33.28%

Peak Wait Times

Saturated intervals: 2.93 → 1.96

+12.98%

Volume Shifted

Away from congested hours via price signalling

+15.42%

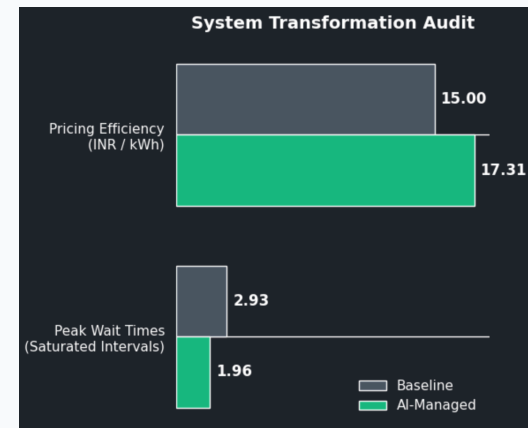
Unit-Level ROI

Efficiency: 15.00 → 17.31 INR/kWh

Physical Grid Relief — Reduced saturated intervals from 2.93 to 1.96, yielding a 33.28% reduction in peak wait times.

Behavioural Load Shaping — Successfully incentivised a 12.98% volume shift away from congested hours via price signalling.

Financial Optimisation — Raised baseline efficiency from 15.00 to 17.31 INR/kWh, generating a 15.42% improvement in unit-level ROI.



Strategic Business & Policy Implications

SLIDE 08

01

Automated Market Making

Operates as an algorithmic market-maker, dynamically adjusting spreads to balance supply and demand in real-time.

02

Infrastructure Deferment

By shaving 33% peak load, operators can safely delay expensive hardware upgrades, maximising current ROI.

03

Grid Policy Alignment

The architecture naturally aligns consumer charging behaviour with macro-grid stability objectives.

Appendix: Model Interpretability

APPENDIX

Feature Importance Analysis

Feature Importance

The model acts as a physics engine, heavily weighting short-term rolling loads over simple time-of-day metrics.

- Rolling 1-hr load momentum is the dominant predictive signal
- Autoregressive lags (t-1, t-2, t-3) outperform static hour-of-day features
- Model learns physical grid dynamics, not calendar patterns

